Gabriel Wan

Internship Availability: Jul 2027 – Dec 2027

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EDUCATION

National University of Singapore **Aug 2025 – Nov 2028 (Expected)**
Bachelor of Computing in Computer Science (Honours), GPA: 4.43/5.00
Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Calculus
Activities: NUS Student Life Hackathon Finalist, NUS Developer Group, IFG Squash Captain

SKILLS

Programming Languages: Python, Java, JavaScript/TypeScript, C, SQL
ML/LLM: PyTorch, Unsloth (LoRA), RAG, pgvector, Slurm
Technologies: FastAPI, Flask, Node.js, React, Next.js, PostgreSQL, Firebase, Docker
Certifications: Google Cloud Professional Machine Learning Engineer

EXPERIENCE

Data Scientist Intern (Incoming) **Jan 2027 – Jun 2027**
Government Technology Agency (GovTech), Cybersecurity Group (CSG)
- Project: Fine-tuning open-weight LLMs (LoRA/QLoRA) for automated red-team payload generation

Digital Transformation Intern **May 2026 – Aug 2026**
Singapore Airlines Engineering Company (SIAEC), Ground Support Equipment (GSE)
- Reduced maintenance record processing from several days to under 1 minute by building a Power Apps/Power Automate/SharePoint workflow adopted by 20 technicians and 2 supervisors
- Automated preventive maintenance scheduling for 800+ GSE by building a Power Automate workflow that sends supervisors monthly overdue and upcoming maintenance reports, replacing manual Excel tracking
- Cut daily battery-check time from 30 minutes to under 1 minute by identifying the underlying telemetry API and building a browser extension that consolidated data from 295 GSE into a searchable dashboard

PROJECTS

KnowBuddy | Python, FastAPI, PostgreSQL, Docker, Tencent Cloud **Oct 2026 – Present**
- Built a permission-aware hybrid RAG system over Slack, Google Drive, Jira and Confluence, combining ParadeDB BM25 and pgvector HNSW search via reciprocal rank fusion under in-query ACL filters
- Secured it with fail-closed live OAuth permission checks per user, PII masking and a tamper-evident hash-chained audit log; prompt-injection defences resisted 30/30 live attacks on Tencent's hy3
- Evaluated on 41 queries against ~21k distractor documents: 0 permission leaks and 36/36 answerable queries answered; p95 search under 450 ms at 100k chunks

mini-gabriel | Python, PyTorch, Unsloth, LoRA, Slurm **Aug 2026 – Present**
- Built an end-to-end pipeline from 14,939 messages: extraction, turn/session segmentation, pseudonymisation, and chat-level train/holdout splitting to prevent data leakage
- Fine-tuned Qwen3-8B with LoRA adapters covering 0.53% of model parameters on an A100 via Slurm, using response-only loss masking verified by decoding training labels before training
- Designed a calibrated style-distance metric (0.3639 to 0.0637, 83% reduction vs. baseline); quantised to 4-bit GGUF (16GB→5GB) and deployed as a CPU-only Telegram bot

KopiKaki | Gemini Live, Next.js, Firebase **Aug 2026**
- Built a voice agent for seniors using Gemini Live, Next.js, and Firebase, turning natural-language requests into confirmed local meetups
- Designed production matching infrastructure with secure Firestore rules, live candidate seeding, and tool calling (Gemini function calling) across meetups, people, groups, and activities

UWash (Finalist, Top 8 — Student Life Hackathon 2026) | Python, Flask, React, ESP32 **Mar 2026**
- Engineered a full-stack IoT laundry monitoring system using ESP32 vibration sensors and a Python Flask backend, enabling real-time machine availability tracking across residential college laundry rooms
- Developed an asynchronous Telegram bot with automated push notifications and a digital queue system, eliminating manual status updates and reducing machine idle time for UTown residential students